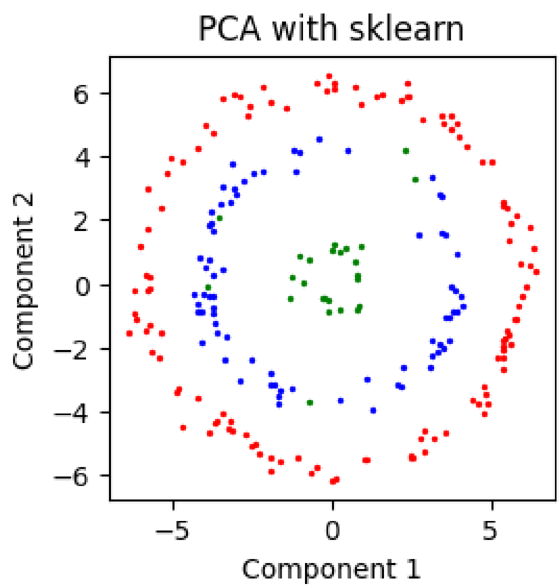
STAT672_Homework1_Bonus

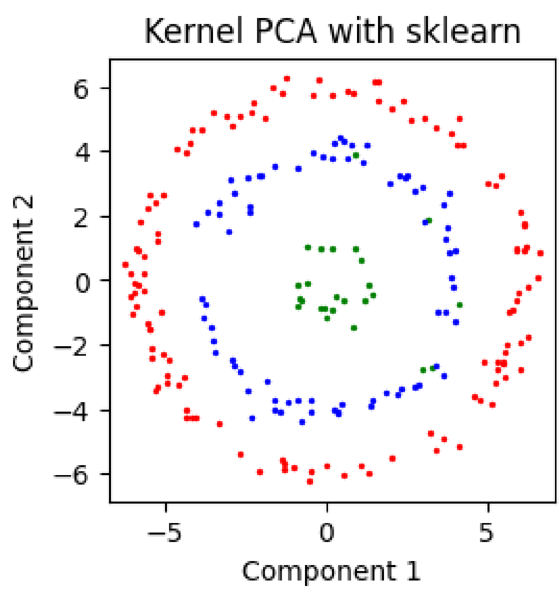
January 27, 2025

[ ]:
```python
#Construct X Matrix from data
X=np.hstack((Xp,Yp))
n_samples = X.shape[0]
start=time.time()
pca=PCA(n_components=2)
PCA=pca.fit_transform(X)
end=time.time()
elapsed_time=(end-start)
inter=[slice(0,int(radius[0]*factor)),slice(int(radius[0]*factor),radius[1]*\
    factor),slice(radius[1]*factor,n_samples)]
for c in range(categories):
    plt.scatter(PCA[inter[c],0],PCA[inter[c],1],s=2,color=color[c])
plt.title("PCA with sklearn")
plt.rcParams['figure.figsize'] = [3, 3]
plt.axis('equal')
plt.xlabel('Component 1')
plt.ylabel('Component 2')
plt.show()
print(f"elapsed time={elapsed_time} seconds")
```



elapsed time=0.0007951259613037109 seconds

[ ]:
```
#Construct X Matrix from data
X=np.hstack((Xp,Yp))
n_samples = X.shape[0]
start=time.time()
kpca=KernelPCA(n_components=2,kernel='linear')
KPCA=pca.fit_transform(X)
end=time.time()
elapsed_time=(end-start)
inter=[slice(0,int(radius[0]*factor)),slice(int(radius[0]*factor),radius[1]*\
    factor),slice(radius[1]*factor,n_samples)]
for c in range(categories):
    plt.scatter(KPCA[inter[c],0],KPCA[inter[c],1],s=2,color=color[c])
plt.title("Kernel PCA with sklearn")
plt.rcParams['figure.figsize'] = [3,3]
plt.axis('equal')
plt.xlabel('Component 1')
plt.ylabel('Component 2')
plt.show()
print(f"elapsed time={elapsed_time} seconds")
```



elapsed time=0.0008051395416259766 seconds